Apollonius problems through GeoGebra

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Abstract

In order to solve the Apollonius problems there will be given GeoGebra code for custom tools. For each problem we review the method used to build a *GeoGebra* tool and finally we reproduce the obtained results.

1 Introduction

GeoGebra [3] is a well known dynamic geometry software package, currently in continuous development. Its purpose is to visualize geometric realities.

In order to solve the Apollonius problems there will be given GeoGebra codes for custom tools. A problem is to find / draw circles tangent to three *things.* The *things* may be points, (straight) lines and circles. The required circles must be tangent to the given circles and / or lines and pass through given points. It results 10 problems:

No.	The problem	Abbr.
1	Circle passing through three points	PPP
2	Circle tangent to three lines	LLL
3	Circle passing through two points and tangent to a line	PPL
4	Circle passing through a point and tangent to two lines	PLL
5	Circle tangent to two lines and a circle	LLC
6	Circle passing through a point and tangent to a line and a circle	PLC
7	Circle tangent to a line and two circles	LCC
8	Circle passing through two points and tangent to a circle	PPC
9	Circle passing through a point and tangent to two circles	PCC
10	Circle tangent to three circles	CCC

The solution of the first problem is the circumscribed circle of the three

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points. The second problem has 4 solutions: the incircle and the three excircles of the triangle defined by the lines. We shall only deal with the rest of the problems and special cases will not be discussed.

The given solutions of the problems are connected. The graph of Fig. 1 presents the dependencies of our codes. Two additional custom tools are used which are presented in an Appendix.

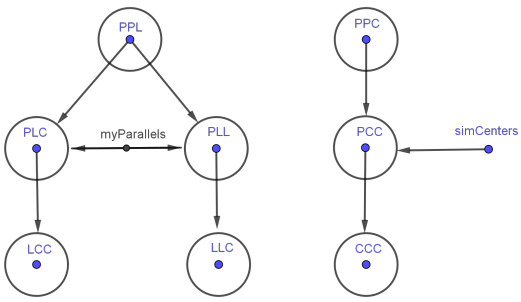

Fig. 1: The graph of dependencies

Same approach is found in [1], [2], but without providing codes for a geometric software. For each problem we review the method used to build a *GeoGebra* tool and finally we reproduce the result of using that tool.

We uses *GeoGebra* 5 (*GeoGebra* classic). With given data the lines of a script must be copied in turn to the *Input* line and executed.

1.1 PPL

Let A and B be two points and the line f, Fig. 3. There is supposed that the lines AB and f are not parallel. If the circle $\mathcal{C}(\Omega)$ is a solution then the power of the point $E = f \cap AB$ w.r. to the circle $\mathcal{C}(\Omega)$ is

$$\wp(E) = |EA| \cdot |EB| = |ET|^2.$$

Knowing T, the circle $\mathcal{C}(\Omega)$ is the circumscribed circle of the points A, B, T. The point T', the symmetry of T with respect of E, gives another solution.

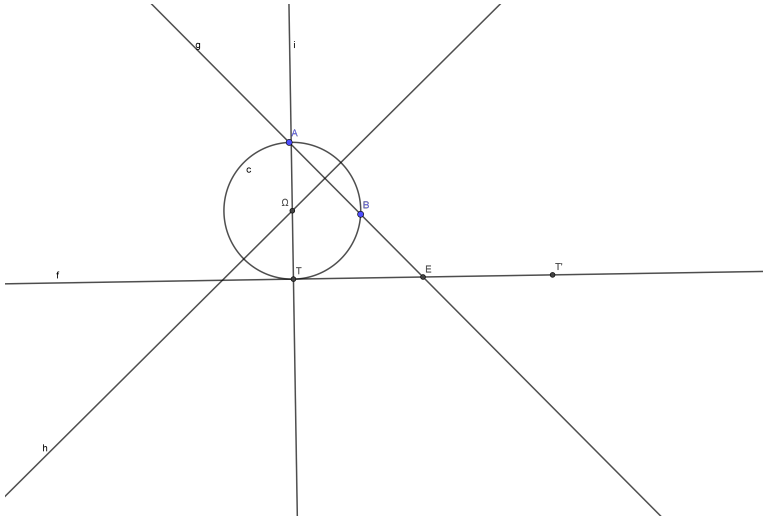

Fig. 2: Reasoning for solving the PPL problem

The *GeoGebra* script is

1	$E = intersect(f, line(A, B))$
2	$T = intersect(f, circle(E, sqrt(segment(A, E) * segment(B, E))))$
3	$c = circle(A, B, T_1)$
4	$d = circle(A, B, T_2)$

If $f = line(C, D)$ then the results of calling $o = PPL(A, B, C, D)$ are given in Fig. 3.

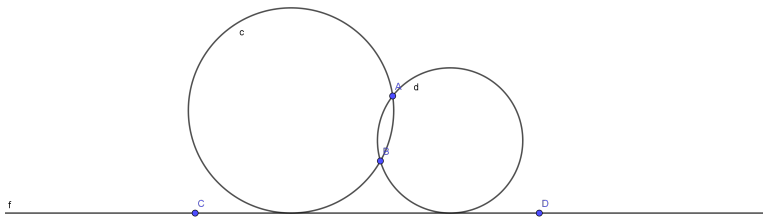

Fig. 3: The results of $o = PPL(A, B, C, D)$

1.2 PLL

Let be the lines AB, AC and the point D, within the angle $\widehat{BAC}$, but not on angle bisector. The center of any solution must be on the angle bisector. If E is the symmetric of D with respect the angle bisector then the solutions are the circles passing through the points D and E and tangent to the line AB, Fig. 4. Thus the problem PLL is reduced to PPL.

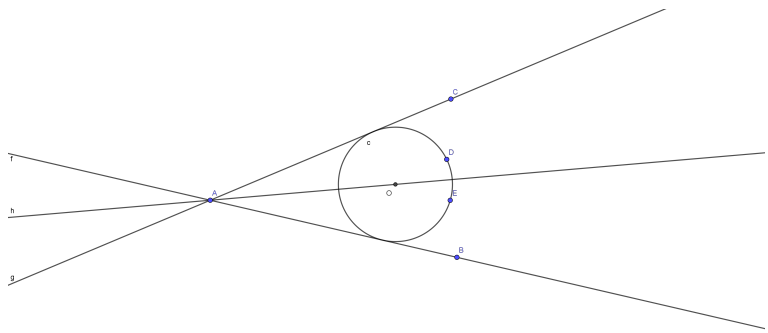

Fig. 4: Reasoning for solving the PLL problem

The *GeoGebra* script is

1	$h = anglebisector(B, A, C)$
2	$E = reflect(D, h)$
3	$o = PPL(D, E, A, B)$

with the results presented in Fig. 5. For the *GeoGebra* tool generated by above script, the calling is $PLL(A, B, C, D)$.

1.3 LLC

The method used here to solve this problem is called the Viète method. We suppose that the circle $\mathcal{C}(D, r), r = |DE|$, is within the angle $\widehat{BAC}$, Fig. 6. We draw the parallels i and j to the lines AB and respectively AC, outside the angle $\widehat{BAC}$ so that the distance between two parallel lines is r. We can construct a circle passing through D and tangent to the lines i and j, (PLL). If that circle is $p = \mathcal{C}(Q, \rho)$ then the circle $q = \mathcal{C}(Q, \rho - r)$ is a solution of the LLC problem.

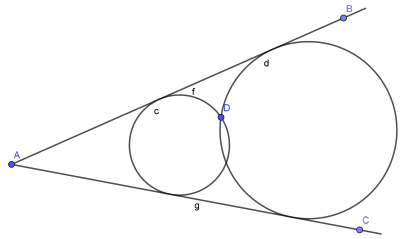

Fig. 5: Results given by the PLL tool

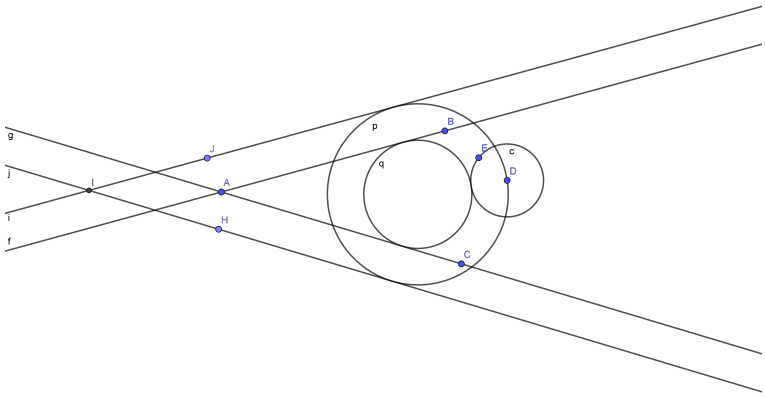

Fig. 6: Reasoning for solving the LLC problem

Similarly, other solutions may be obtained for the parallels that intersect the interior of the angle $\widehat{BAC}$.

To construct two parallels to a line at a given distance, the *myParallels* tool, A.1, is used in the following script:

1	$i = myParallels(A, B, radius(c))$
2	$j = myParallels(A, C, radius(c))$
3	$I_1 = intersect(i_1, j_2)$
4	$I_2 = intersect(i_2, j_1)$
5	$J_1 = point(i_1)$
6	$H_1 = point(j_2)$
7	$o1 = PLL(I_1, J_1, H_1, D)$
8	$J_2 = point(i_2)$
9	$H_2 = point(j_1)$
10	$o2 = PLL(I_2, J_2, H_2, D)$
11	$p_1 = circle(center(o1_1), radius(o1_1) - radius(c))$
12	$p_2 = circle(center(o1_2), radius(o1_2) - radius(c))$
13	$q_1 = circle(center(o2_1), radius(o2_1) + radius(c))$
14	$q_2 = circle(center(o2_2), radius(o2_2) + radius(c))$

The results of the above codes are given in Fig. 7. It is important that the order of the points B and C are given in such a way that the ray AB rotates over the ray AC counterclockwise and $\widehat{BAC} < 180^o$. For the example of Fig. 7, the calling of the *GeoGebra* tool will be $LLC(A, C, B, D, E)$.

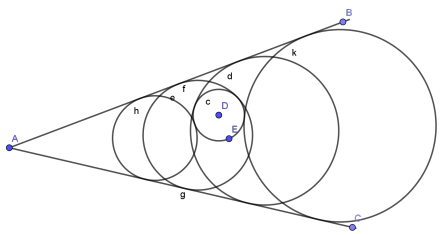

Fig. 7: Results of the LLC tool

1.4 PLC

There are given (see Fig. 8):

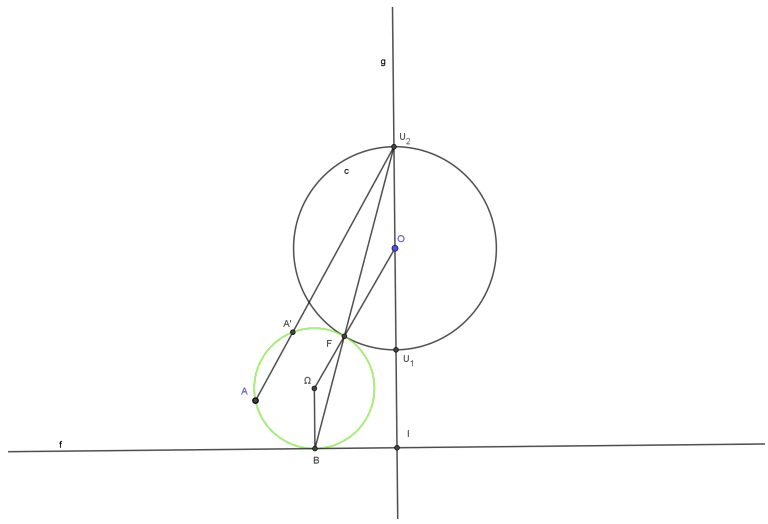

Fig. 8: Reasoning image for solving the PLC problem

1. the point A;
2. the line f;
3. the circle c with the center O. The circle and the point A are on the same side of the line f.

Let I be the intersection between f and the perpendicular through O to f, while U_1, U_2 are the intersections between that perpendicular with the circle c.

We suppose that the circle $\mathcal{C}(\Omega)$, with center Ω, is a solution of the PLC problem. Denoting by F the intersection point between c and $\mathcal{C}(\Omega)$, it results that the point O, F and Ω are collinear. Let be $\Omega B \perp f,\ B \in f$.

We use the inversion with center U_2 that transforms the circle c into the line f. Then I is the images of U_1 through this inversion.

We observe that

$$U_2 I /\!/ \Omega B \quad \Rightarrow \quad \widehat{B\Omega F} = \widehat{U_2 O F} \Rightarrow \quad \widehat{\Omega F B} = \widehat{U_2 F O}$$

It results that the points U_2, F and B are collinear points and B is the image of F through inversion.

Let A' be the second intersection point between the line AU_2 with the circle $\mathcal{C}(\Omega)$. The power of point U_2 w.r. to the circle $\mathcal{C}(\Omega)$ gives

$$\wp(U_2) = |U_2 A| \cdot |U_2 A'| = |U_2 B| \cdot |U_2 F|.$$

It results that the points A and A' are inverse to each other. Moreover the inversion preserves the circle $\mathcal{C}(\Omega)$.

From the equality $|U_2A| \cdot |U_2A'| = |U_2U_1| \cdot |U_2I|$ it results that the points A', U_1, I, A are concyclical. We may obtain A' as the intersection between U_2A with the circle passing through A, I, U_1. Thus the PLC problem is reduced to the PPL problem: find the circles passing through A, A' and tangent to the line f.

Other solutions are obtained taking into account the inversion with center U_1 transforming the circle c into the line f.

The *GeoGebra* script is

1	$I = intersect(f, perpendicularline(D, f))$
2	$U = intersect(c, perpendicularline(D, f))$
3	$F = intersect(line(A, U_2), circle(A, I, U_1))$
4	$G = intersect(line(A, U_1), circle(A, I, U_2))$
5	$o1 = PPL(A, F_2, B, C)$
6	$o2 = PPL(A, G_2, B, C)$

Calling PLC(A, B, C, D, E) the results are given in Fig. 9.

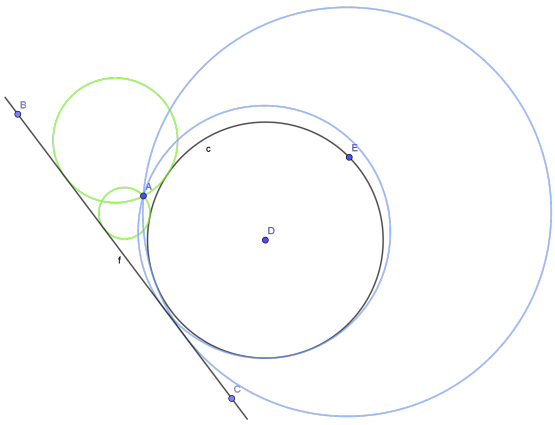

Fig. 9: Results of the PLC tool

1.5 LCC

The Viète method will be used. Given the line $f = line(A, B)$ and the circles $c = \mathcal{C}(C, r_1)$, $d = \mathcal{C}(D, r_2)$ with $r_1 > r_2$, Fig. 10, we draw parallels

to f and the circle $c_1 = \mathcal{C}(C, r1 - r2)$. If the circle $\mathcal{C}(\Omega, \rho)$ is a solution then the circle $\mathcal{C}(\Omega, \rho + r_2)$ passes through the point E and is tangent to the line l_1 and the circle c_1. The LCC problem is reduced to a PLC problem.

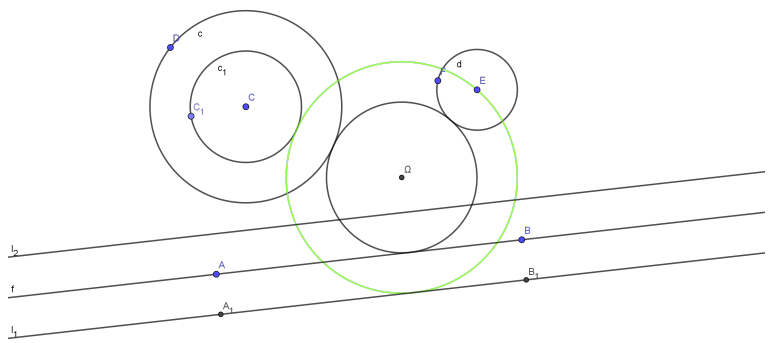

Fig. 10: Reasoning image for solving the LCC problem

Taking into account all the solutions of the PLC problem it results the *GeoGebra* script

1	$l = myParallels(f, radius(d))$
2	$c_1 = circle(C, radius(c) - radius(d))$
3	$C_1 = point(c_1)$
4	$A_1 = intersect(l_1, perpendicularline(A, f))$
5	$B_1 = intersect(l_1, perpendicularline(B, f))$
6	$o1 = PLC(E, A_1, B_1, C, C_1)$
7	$e1 = circle(Center(o1_2), radius(o1_2) - radius(d))$
8	$g1 = circle(Center(o1_1), radius(o1_1) - radius(d))$
9	$c_2 = circle(C, radius(c) + radius(d))$
10	$C_2 = point(c_2)$
11	$A_2 = intersect(l_2, perpendicularline(A, f))$
12	$B_2 = intersect(l_2, perpendicularline(B, f))$
13	$o2 = PLC(E, A_2, B_2, C, C_2)$
14	$e2 = circle(center(o2_1), radius(o2_1) + radius(d))$
15	$g2 = circle(center(o2_2), radius(o2_2) + radius(d))$

The four solution of LCC(A, B, C, D, E, F) are given in Fig. 11.

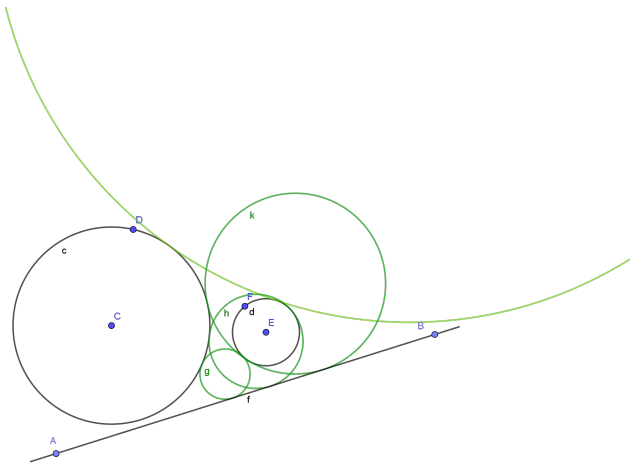

Fig. 11: Result of LCC(A, B, C, D, E, F)

1.6 PPC

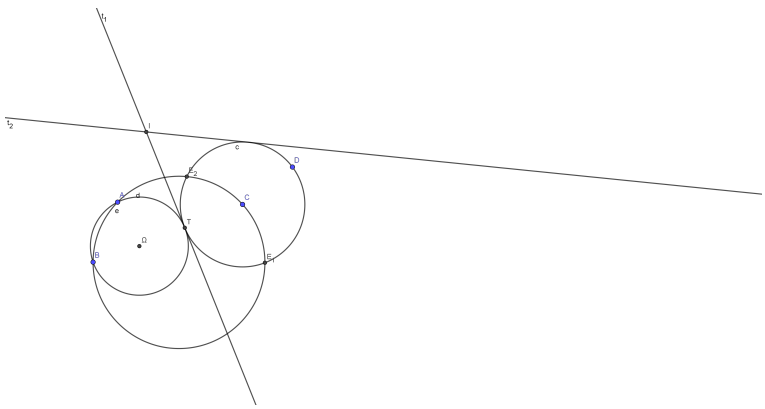

Fig. 12: Reasoning image for solving the PPC problem

Let A and B be two points and the circle $c = \mathcal{C}(C)$, Fig. 12. The circle passing through the points A, B, C intersects c in the points E_1, E_2. There is supposed that the lines AB and E_1E_2 are not parallel, case that occurs when the point C is on the perpendicular bisector of segment AB.

If $\mathcal{C}(\Omega)$ is a solution of the PPC problem then the lines AB, E_1E_2 and the exterior tangents to the circle c intersect in I, the radical center of $c, \mathcal{C}(\Omega)$ and $\mathcal{C}(A, B, C)$.

If T is the intersection between an exterior tangent and c then $\mathcal{C}(\Omega)$ is the circle passing through A, B and T. It results the following script

1	$d = circle(A, B, C)$
2	$E = intersect(c, d)$
3	$I = intersect(line(E_1, E_2), line(A, B))$
4	$t = tangent(I, c)$
5	$e = circle(intersect(t_1, c), A, B)$
6	$g = circle(intersect(t_2, c), A, B)$

The results of the PPC tool are given in Fig. 13.

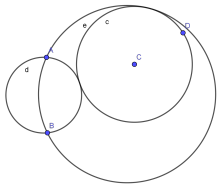

Fig. 13: Results of calling PPC(A, B, C, D)

1.7 PCC

There are given the point A and the circles $\mathcal{C}(O_1, r_1)$ and $\mathcal{C}(O_2, r_2)$, Fig. 14. Let be S_e the exterior similarity center and suppose that the circle $\mathcal{C}(\Omega)$ is a solution to the PCC problem. We denote by

- T_1 and T_2 the tangent points between $\mathcal{C}(\Omega)$ and the given circles.
- B the second intersection between the line AS_e and $\mathcal{C}(\Omega)$.

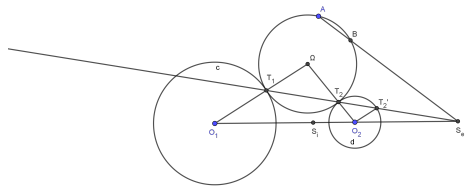

Fig. 14: Reasoning image for solving the PCC problem

The points T_1, T_2, S_e are collinear, (see A.2).

Computing the power of S_e w.r. to the circle $\mathcal{C}(\Omega)$ we get from A.2(1)

$$\wp(S_e) = (|S_eO_2|^2 - r_2^2) \cdot \frac{r_1}{r_2}$$

and also

$$\wp(S_e) = |S_eA| \cdot |S_eB|.$$

From the above two equalities it results the position of the point B

$$|S_eB| = (|S_eO_2|^2 - r_2^2) \cdot \frac{r_1}{r_2 \cdot |S_eA|}$$

Of cource it results two possible solutions for the point B. In this way the PCC problem is reduced to the PPC problem: construct the circles passing through the points A, B and tangent to circle $\mathcal{C}(O_1, r_1)$.

Considering the interior similarity center it results other solutions of the PCC problem.

The *GeoGebra* script is

1	$S = simCenters(B, C, D, E)$
2	$r1 = radius(c)$
3	$r2 = radius(d)$
4	$s1 = (segment(S_1, D)^2 - r2^2) * r1/(r2 * segment(S_1, A))$
5	$F_1 = intersect(ray(S_1, A), circle(S_1, s1))$
6	$o1 = PPC(A, F_1, B, C)$
7	$s2 = (segment(S_2, D)^2 - r2^2) * r1/(r2 * segment(S_2, A))$
8	$A' = reflect(A, S_2)$
9	$F_2 = intersect(ray(S_2, A'), circle(S_2, s2))$
10	$o2 = PPC(A, F_2, B, C)$

Calling the corresponding *GeoGebra* tool the results are given in Fig. 15.

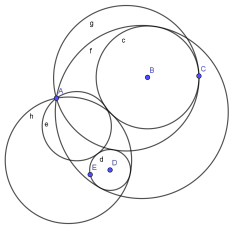

Fig. 15: Results of calling PCC(A, B, C, D, E)

1.8 CCC

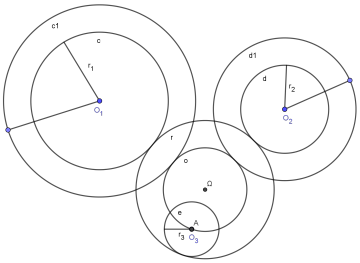

Fig. 16: Reasoning image for solving the CCC problem

There are given three circles $\mathcal{C}(O_i, r_i),\ i \in \{1, 2, 3\}$ with $r_1 > r_2 > r_3$, Fig. 16. Using the Viète method, if $\mathcal{C}(\Omega, \rho)$ is a solution then circle $\mathcal{C}(\Omega, \rho - r_3)$ passes through the point O_3 and are tangent to the circles $\mathcal{C}(O_i, r_i + r_3),\ i \in \{1, 2\}$. Thus the problem CCC is reduced to the problem PCC.

The solutions will be obtained solving four PCC problems, generate the circles passing through O_3 and tangent to circles $\mathcal{C}(O_1, r_1 \pm r_3)$ and $\mathcal{C}(O_2, r_2 \pm r_3)$, considering all combinations of + and - operations.

The corresponding *GeoGebra* script is

1	$c1 = circle(A, radius(c) + radius(e))$
2	$d1 = circle(C, radius(d) + radius(e))$
3	$B1 = point(c1)$
4	$D1 = point(d1))$
5	$o1 = PCC(E, A, B1, C, D1)$
6	$q1 = circle(center(o1_2), radius(o1_2) - radius(e))$
7	$c2 = circle(A, radius(c) - radius(e))$
8	$d2 = circle(C, radius(d) - radius(e))$
9	$B2 = point(c2)$
10	$D2 = point(d2)$
10	$o2 = PCC(E, A, B2, C, D2)$
11	$q2 = circle(center(o2_1), radius(o2_1) - radius(e))$
12	$r2 = circle(center(o2_2), radius(o2_2) + radius(e))$
13	$o3 = PCC(E, A, B1, C, D2)$
14	$q3 = circle(center(o3_3), radius(o3_3) + radius(e))$
15	$r3 = circle(center(o3_4), radius(o3_4) - radius(e))$
16	$o4 = PCC(E, A, B2, C, D1)$
17	$q4 = circle(center(o4_3), radius(o4_3) + radius(e))$
18	$r4 = circle(center(o4_4), radius(o4_4) - radius(e))$

The results of the CCC tool are given in Fig. 17.

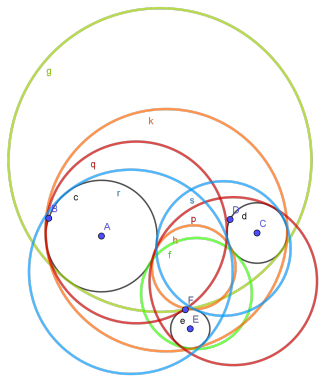

Fig. 17: The results of CCC(A, B, C, D, E, F)

Finally we mention the problem of *Soddy* circles, a particular case of the CCC problem. The *Soddy* problem is to find the circles tangent to three pairwise tangent circles, Fig. 18.

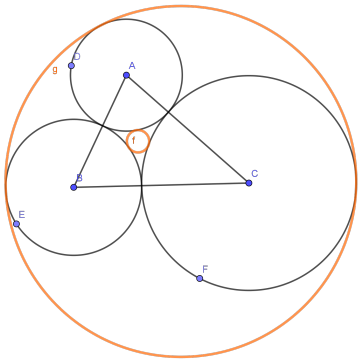

Fig. 18: Soddy circles

2 Conclusions

Instead of using the mouse to pick command from menus we prefer to edit / copy commands in the *Input* line. That's amore programmable style of using *GeoGebra*. It would be useful for *GeoGebra* to offer the possibility to upload and process a command file.

The *Manage Tool* doesn't allow the possibility to upload tools. If a construction needs two custom tools then we shall start with a file with that tools, created successively, that's an annoying procedure.

A Additional tools

A.1 Parallels to a line at a given distance (myParallels)

There are given the line $f = line(A, B)$ and a positive number r. The *GeoGebra* script that generates two parallel line to f at the distance r is

1	$F = intersect(perpendicularline(A, f), circle(A, r))$
2	$l_1 = line(F_1, f)$
3	$l_1 = line(F_1, f)$

The results using the corresponding tool are presented in Fig. 19.

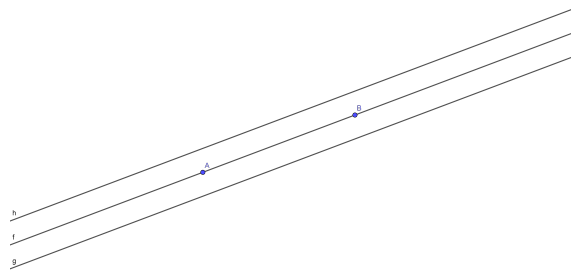

Fig. 19: Results of myParallels$(A, B, 1)$

A.2 Similarity centers (simCenters)

Two circles $\mathcal{C}(O_1, r_1), \mathcal{C}(O_2, r_2)$, not included one in other, have an exterior similarity center. If the circles doesn't intersect then there exists an interior similarity center, too. In Fig. 20, the point S_e is the exterior similarity center, while S_i is the interior similarity center.

If $r_1 > r_2$ then the following relations hold

$$|S_eO_2| = \frac{r_2}{r_1 - r_2}|O_1O_2|, \qquad |S_iO_2| = \frac{r_2}{r_1 + r_2}|O_1O_2|.$$

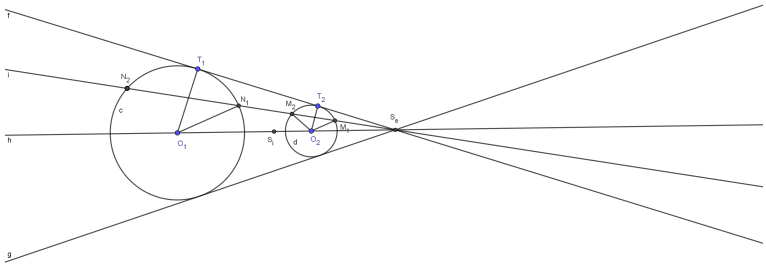

Fig. 20: The similarity centers

Based on the above relations, the *GeoGebra* script to fix the similarity centers is

1	$f = line(A, C)$
2	$r1 = radius(c)$
3	$r2 = radius(d))$
4	$re = min(r1, r2)/abs(r1 - r2)$
5	$E = intersect(f, circle(if(r1 > r2, C, A), re * segment(A, C))))$
6	$Se = if(r1 > r2, E_2, E_1)$
7	$ri = r2/(r1 + r2)$
8	$F = intersect(f, circle(C, ri * segment(A, C)))$
9	$Si = F_1$

The results of calling simCenters(A, B, C, D) are given in Fig. 21.

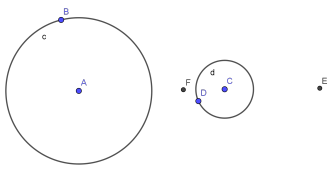

Fig. 21: The results of simCenters tool

We establish some properties related to the similarity centers:

- If the line i intersect the circles c and d in the points N_1, N_2 and respectively M_1, M_2, Fig. 20, then the triangles $\triangle S_e O_1 N_1$ and $\triangle S_e O_2 M_2$ are similaries. As a consequence it results that $O_1N_1 /\!\!/ O_2M_1$. A similar result occurs in the case of interior similarity center.

- If a circle $\mathcal{C}(\Omega, \rho)$ is tangent to the circles $\mathcal{C}(O_i, r_i),\ i \in \{1, 2\}$ with the tangent points T_1, T_2, Fig. 14, then the points T_1, T_2, S_e are colinear. This may be proved using the Menelaus's Theorem in $\triangle OO_1O_2$.

- Computing the power of S_e w.r. to the circle $\mathcal{C}(\Omega)$, Fig. 14, we obtain

$$\wp(S_e) = |S_eT_1| \cdot |S_eT_2| = |S_eT_2| \cdot |S_eT_2'| \cdot \frac{|S_eT_1|}{|S_eT_2'|} =$$

$$= wp_{\mathcal{C}(O_2)}(S_e) \cdot \frac{|S_eT_1|}{|S_eT_2'|} = (|S_eO_2|^2 - r_2^2) \cdot \frac{r_1}{r_2} = |S_eO_1| \cdot |S_eO_2| - r_1r_2. \quad (1)$$

  There is another meaning of the above equality. If we consider the inversion with center S_e mapping the circle $\mathcal{C}(O_2, r_2)$ onto $\mathcal{C}(O_1, r_1)$ then the power of inversion is

$$\rho^2 = |S_eT_1| \cdot |S_eT_2| = |S_eO_1| \cdot |S_eO_2| - r_1r_2.$$

References

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